Stock Data Intelligence Dashboard

A Mini Financial Data Platform — Real-time NSE Stock Analytics, REST APIs & Interactive Visualizations

Python · FastAPI · Pandas · yfinance · Chart.js

2026

Project Objective & Overview

Data Collection

Fetch 2 years of real NSE stock data using yfinance for 10 major companies, clean & transform with Pandas.

REST API Layer

5 FastAPI endpoints expose OHLCV data, summaries, comparisons & market movers with auto-generated Swagger docs.

Visual Dashboard

Interactive browser dashboard with Chart.js — price charts, MA overlays, KPI cards & stock comparison tool.

Evaluated: Python & Data (30%) · API Design (25%) · Creativity (15%) · Visualization (15%) · Docs (10%) · Bonus (5%)

Technology Stack

Language	**Python 3.11+**	Core language for backend, data processing & APIs
Framework	**FastAPI + Uvicorn**	High-performance async REST API with auto Swagger docs
Data	**Pandas + NumPy**	Data cleaning, transformation & metric computation
Source	**yfinance**	Free real-time NSE/BSE stock data, no API key needed
Frontend	**HTML + JS + Chart.js**	Zero-dependency interactive dashboard, no build step
Cache	**Local CSV (6h TTL)**	Avoids rate limits; auto-refreshes stale data on request

Data Pipeline & Metrics

Fetch via yfinance → Clean & Validate → Compute Metrics → Cache as CSV → Serve via API

Computed Metrics

Daily Return

(CLOSE - OPEN) / OPEN x 100

7-Day MA

Rolling mean of Close (7 days)

30-Day MA

Rolling mean of Close (30 days)

52W High / Low

Max High / Min Low over 252 sessions

Volatility Score

Std Dev of Daily Returns x 100

Momentum Score

(30d avg - 90d avg) / 90d avg x 100

REST API Endpoints

GET	`/companies` List all 10 NSE companies with names	Core
GET	`/data/{symbol}` OHLCV + Daily Return + MA-7 + MA-30 (last N days)	Core
GET	`/summary/{symbol}` 52W High/Low, Avg Close, Volatility, Momentum	Core
GET	`/compare` Normalized chart + Pearson correlation + returns	Bonus
GET	`/market/movers` Top 5 Gainers and Top 5 Losers by daily change	Bonus

Full interactive Swagger UI at http://localhost:8000/docs

Interactive Dashboard Features

Company Sidebar

All 10 NSE companies with live daily % change (green/red)

Price Chart

Closing price + MA-7 + MA-30 overlays using Chart.js

KPI Summary Cards

6 live cards: Price, Daily Change, 52W High/Low, Avg, Volatility, Momentum

Time Filters

Switch between 30D / 90D / 6M / 1Y with one click

Stock Comparison

Normalized base-100 dual-line chart with Pearson correlation

Market Movers

Top 5 Gainers and Losers banner — click to load any stock

Custom Metrics — Creativity & Insight

Volatility Score

Formula:

stddev(daily returns) x 100

What it measures:

Measures how wildly a stock swings day to day.

Key Insight:

High = risky, Low = stable. 1.5% means avg daily deviation is 1.5%.

Momentum Score

Formula:

((MA30 - MA90) / MA90) x 100

What it measures:

Compares recent trend vs longer-term average.

Key Insight:

Positive = bullish momentum. Negative = bearish signal.

Pearson Correlation

Formula:

corr(stock1_close, stock2_close)

What it measures:

Quantifies how two stocks move together.

Key Insight:

Range -1 to +1. >0.8 = highly correlated. <0 = inverse (diversification).

Live Data Snapshot — NSE Stocks

Symbol	Price (INR)	Daily Chg	52W High	52W Low	Momentum
RELIANCE	1,345	-0.18%	1,611	1,110	-4.6%
TCS	2,368	-0.88%	3,586	2,348	-13.7%
INFY	1,259	-0.80%	1,975	1,148	-10.2%
HDFCBANK	744	-1.49%	1,880	1,364	-8.4%
WIPRO	190	-0.52%	580	458	-5.1%
ICICIBANK	1,214	-1.53%	1,488	1,206	-2.7%
BAJFINANCE	818	-3.01%	7,830	6,187	-15.2%

Data fetched live from Yahoo Finance via yfinance · Refreshed every 6 hours from local CSV cache

Project Structure & Architecture

```
stock_dashboard/
  backend/
    main.py
    data_loader.py
  frontend/
    index.html
  data/   (CSV cache)
  requirements.txt
  README.md
```

main.py

FastAPI app with 5 endpoints, CORS, static serving, NaN-safe JSON responses

data_loader.py

yfinance fetch → Pandas transform → CSV cache (6h TTL) → derived columns

index.html

Single-file frontend with Chart.js charts, no build step required

data/ folder

Auto-created CSV cache; one file per symbol, refreshed when stale

Project Summary

Stock Data Intelligence Dashboard — JarNox Internship Assignment

Part 1 - Real NSE data via yfinance + Pandas cleaning + 6 computed metrics

Part 2 - 5 FastAPI endpoints + Swagger UI auto-docs + NaN-safe JSON

Part 3 - Interactive Chart.js dashboard with filters, KPI cards & compare tool

Bonus - Market Movers, Pearson Correlation, Momentum Score, Volatility Score

Docs - README with setup, API reference, design decisions

Submit to: support@jarnox.com | Include GitHub link + README

Built with Python · FastAPI · Pandas · yfinance · Chart.js